

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being — HDI Prediction using AI/ML
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal report aims to simplify HDI prediction using machine learning, boosting accessibility and accuracy. It addresses the complexity of manual HDI computation, promising faster analysis, reduced errors, and actionable development insights for researchers, students, and policy analysts. Key features include a machine learning-based HDI prediction model, an HDI category classifier, and an interactive dashboard suite for data-driven exploration.

Project Overview	
Objective	The primary objective is to predict the Human Development Index (HDI) of a country using machine learning, enabling researchers, students, and policy analysts to instantly assess human development levels based on key socioeconomic indicators — Life Expectancy, Mean Years of Schooling, GNI per capita, and Internet Users.
Scope	The project comprehensively covers data collection, preprocessing, model training, and web-based deployment. It incorporates a Linear Regression ML model served via a Flask application, with three interactive dashboards extending the analytical scope beyond basic prediction.
Problem Statement	
Description	Manually computing HDI scores requires access to multiple datasets and complex calculations across three dimensions (health, education, and income). This process is time-consuming, error-prone, and inaccessible to non-technical users who need quick development insights.
Impact	Solving this problem enables instant, data-driven HDI estimation for any custom combination of indicators, supporting faster policy analysis, academic research, and educational understanding of global development disparities.
Proposed Solution	
Approach	Employing supervised machine learning (Linear Regression) trained on the Kaggle HDI dataset to build a prediction model that takes socioeconomic indicators as input and outputs a precise HDI score with a classified category (Low / Medium / High / Very High HDI).
Key Features	- HDI Score Prediction using Linear Regression model (HDI.pkl) - HDI Category Classification into four development levels - Data preprocessing pipeline (null handling, feature selection, train/test split) - Flask web application with input form

	and result display - HDI Score Comparison Dashboard (Top 20 / Bottom 10 countries) - World HDI Map Dashboard (filterable by category) - HDI Trend Over Years Dashboard (1990–2015 by region)
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications and development machine	Intel Core i5 / i7 or equivalent (local development machine)
Memory	RAM specifications	8 GB RAM (minimum)
Storage	Disk space for data, model files, and project	~500 MB (dataset + model + templates)
Software		
Frameworks	Python web framework	Flask 3.0.3
Libraries	ML and data analysis libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE and notebook	VS Code, Jupyter Notebook, Anaconda
Data		
Dataset	Source, size, format	Kaggle HDI Dataset — 195 rows × 82 columns, CSV format (HDI.csv, Continent_HDI.csv, countries.geojson)